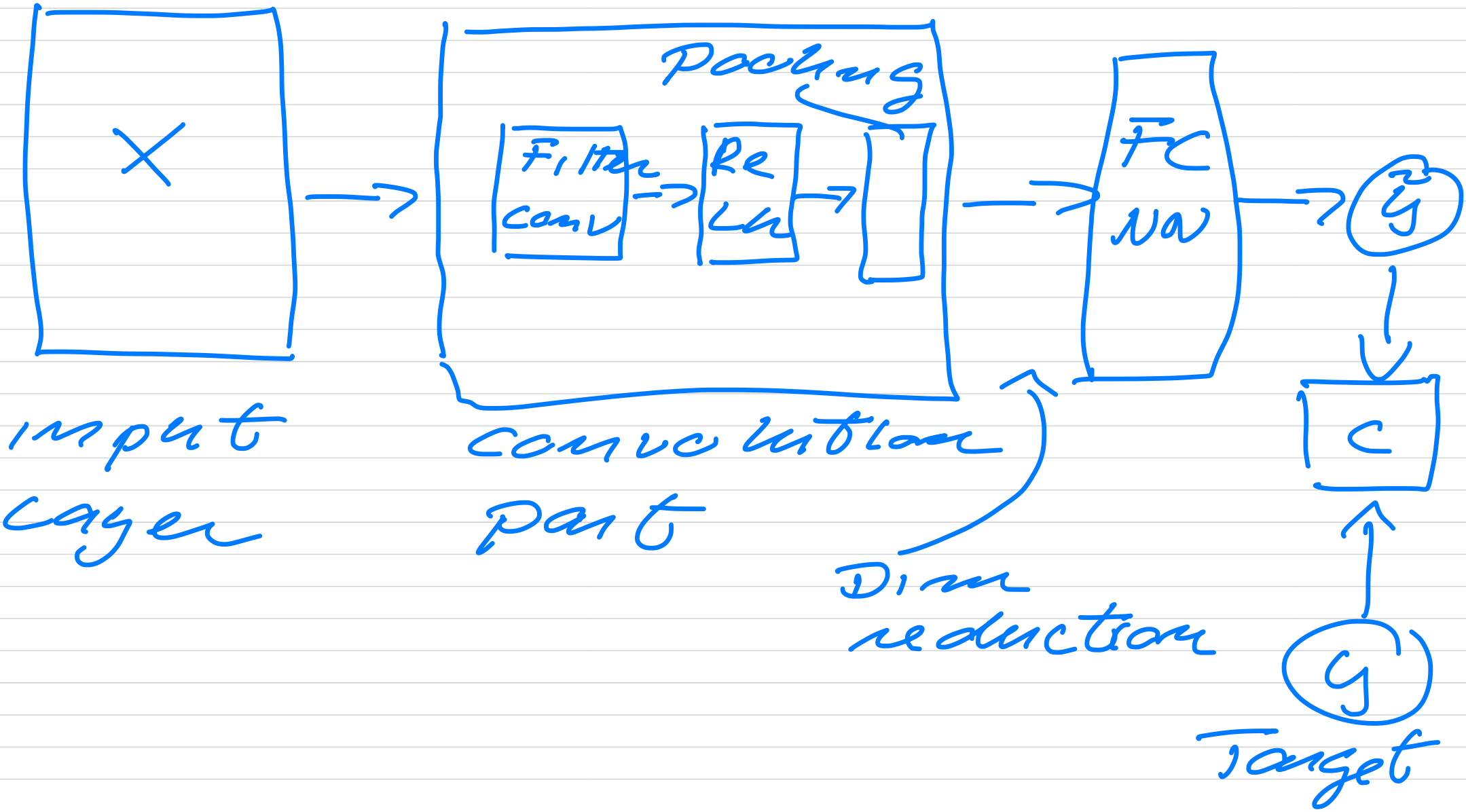
FYS-STK3155/4155 October 26

FYS-STK3155/4155 October 26



$$p(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2$$

$$s(t) = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 t^3$$

$$z(t) = p(t) \cdot s(t)$$

$$= \delta_0 + \delta_1 t + \delta_2 t^2 + \delta_3 t^3 + \delta_4 t^4 + \delta_5 t^5$$

$$\delta_0 = \alpha_0 \beta_0 \; ; \quad \delta_1 = \alpha_1 \beta_0 + \alpha_0 \beta_1$$

$$\delta_2 = \alpha_0 \beta_2 + \alpha_1 \beta_1 + \alpha_2 \beta_0$$

$$\delta_3 = \alpha_1 \beta_2 + \alpha_2 \beta_1 + \alpha_0 \beta_3 \; ; \quad \delta_4 = \alpha_2 \beta_2 + \alpha_1 \beta_3$$

$$\delta_5 = \alpha_2 \beta_3$$

$$\alpha_i' = 0 \quad \text{except} \quad i = \{0, 1, 2\}$$

$$\beta_i' = 0 \quad \text{--}\iota\text{--} \quad i = \{0, 1, 2, 3\}$$

$$S_j = \sum_{i=-\infty}^{\infty} \alpha_i' \beta_{j-i}' = (\alpha * \beta)_j$$

$$S = \begin{bmatrix} \beta_0 & 0 & 0 \\ \beta_1 & \beta_0 & 0 \\ \beta_2 & \beta_1 & \beta_0 \\ \beta_3 & \beta_2 & \beta_1 \\ 0 & \beta_3 & \beta_2 \\ 0 & 0 & \beta_3 \end{bmatrix} \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{bmatrix}$$

← Toeplitz

replace R with x (input)

— " — α — " — w (weights of filter)

— " — δ — " — y (output)

$$y(i) = (x * w)(i)$$

$$= \sum_{k=0}^{k=m-1} w(k)\, x(i-k)$$

$m = 3$ $\quad w = \{ \underset{\alpha_0}{w(0)}, \underset{\alpha_1}{w(1)}, \underset{\alpha_2}{w(2)} \}$

$x = \{ x(0), x(1), x(2), x(3) \}$

For specific i-value we can have $x(-1)$ or $(x-2)$ which are not defined in X

Add two x-value

$x(-2), x(-1), x(0), x(1), x(2)$

$c \quad 0 \quad \beta_0 \quad \beta_1 \quad \beta_2$

$x(3), x(4), x(5)$ enlarge

$\beta_3 \quad c \quad 0$ by $2p$

$p = 2$

P = Padding (hyperparameter, fixed)

$$X = \begin{bmatrix} x_{00} & x_{01} & x_{02} \\ x_{10} & x_{11} & x_{12} \\ x_{20} & x_{21} & x_{22} \end{bmatrix}$$

S = Stride = 1

$$W = \begin{bmatrix} w_{00} & w_{01} \\ w_{10} & w_{11} \end{bmatrix}$$

$$Y = \begin{bmatrix} x_{00}w_{00} + x_{01}w_{01} + x_{10}w_{10} + x_{11}w_{11} & x_{01}w_{00} + x_{02}w_{01} + x_{11}w_{10} + x_{12}w_{11} \\ x_{10}w_{00} + x_{11}w_{01} + x_{20}w_{10} + x_{21}w_{11} & x_{11}w_{00} + x_{12}w_{01} + x_{21}w_{10} + x_{22}w_{11} \end{bmatrix}$$

Padding : $P = 0$

$P = 1$

0	0	0	0	0
0	x_{00}	x_{01}	x_{02}	0
0	x_{10}	x_{11}	x_{12}	0
0	x_{20}	x_{21}	x_{22}	0
0	0	0	0	0